Midterm Skills Exam	
Course Code: CPE312	**Program: CpE**
Course Title: Predictive Analytics using Machine Learning	**Date Performed: 9/24/26**
Section: CP31S3	**Date Submitted: 9/24/26**
Name: Abon, Benedict Aldous A.	**Instructor: Engr. Robin Valenzuela**

1. Discussion

The activity is composed on the Midterm Skills Exam about the implementation of different supervised learning models on fetal health. These models are then evaluated through accuracy, precision, recall, ROC curve, and confusion matrices.

2. Materials and Equipment

- Google Colab

3. Procedure

Use the given dataset and perform the following tasks fetal_health.csv

Please include your pre-processing steps.

Item No. 1: Train and build a machine learning model using the following ML algorithms. Make sure to hyper-tune the parameters of the ML algorithm.

- Random Forest
- Decision Trees
- Bagging
- Stacked Models

Item No. 2: Use the following to evaluate the performance of the model for each ML algorithm:

- Accuracy
- Precision and Recall
- ROC AUC
- Confusion Matrix

Item No. 3: Compare the performance metrics of the model and interpret your result.

Submission Requirements:

- PDF report (do not print your Python notebook).
- Share your Google Colab link for clarity. If multiple notebooks, submit all accessible links

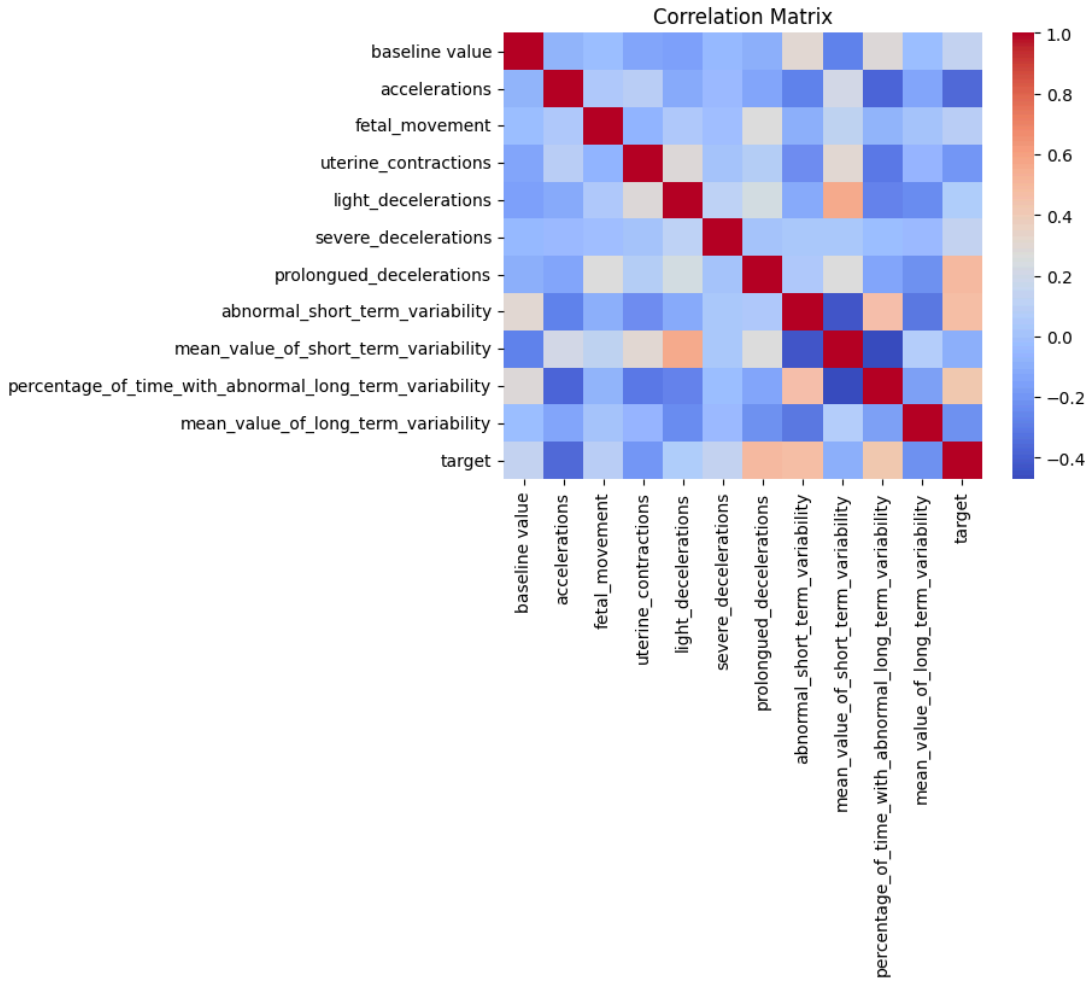
Rules:

1. No using of phones
2. No searching
3. No using of AI tools
4. No using of tools outside of that is allowed (Colab)

Anyone caught using these will have a score of 0

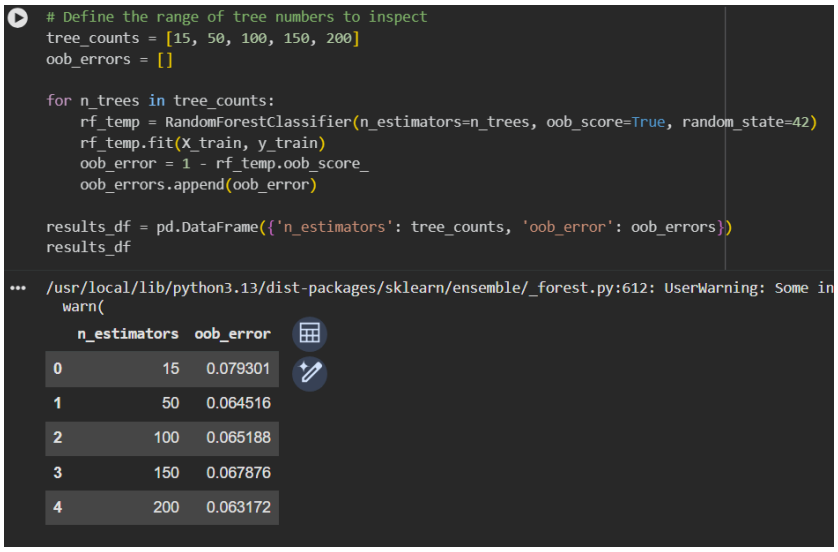
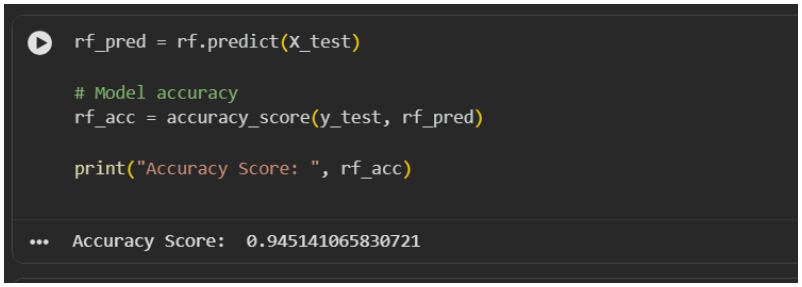
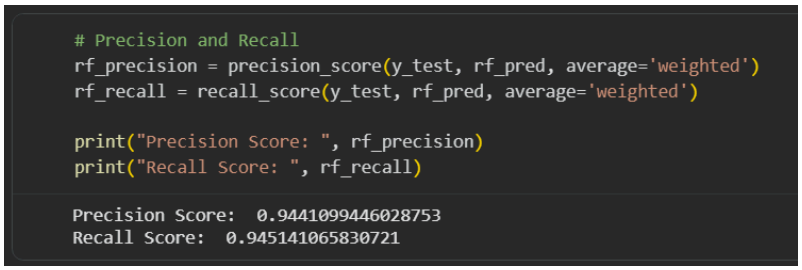
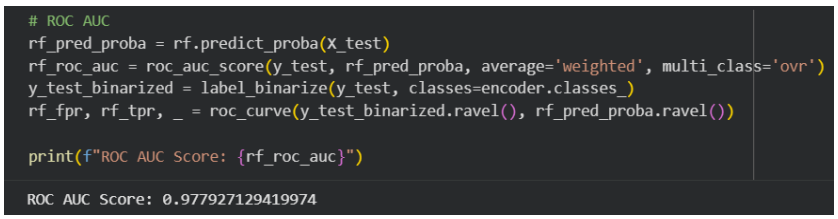
4. Output

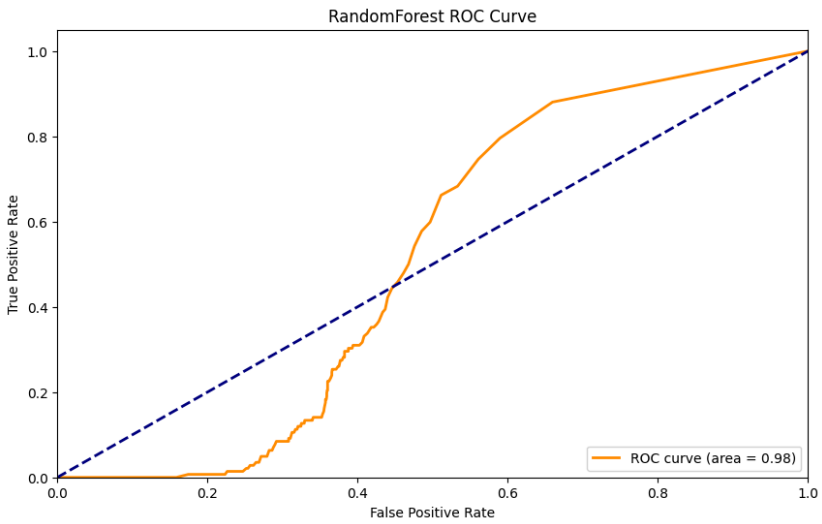
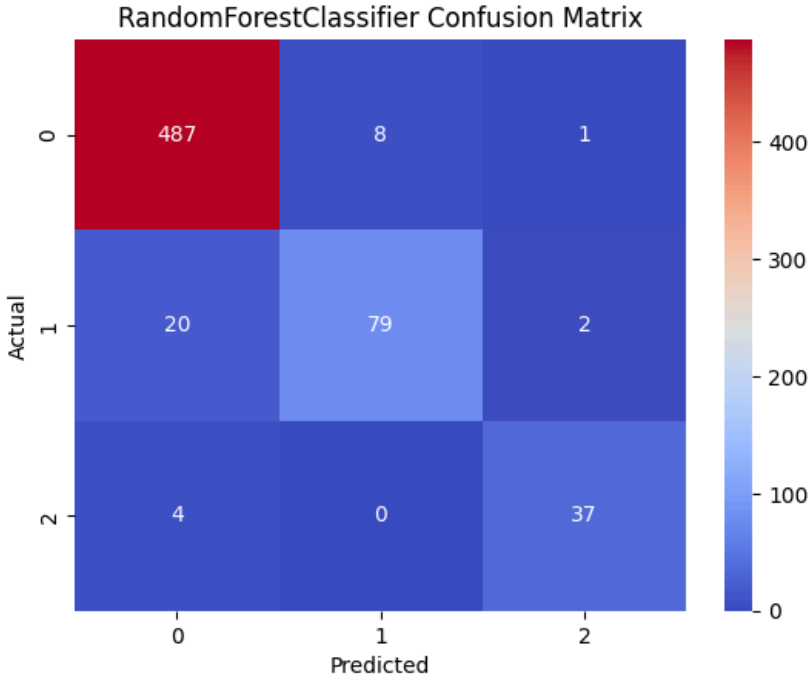
Initial Results

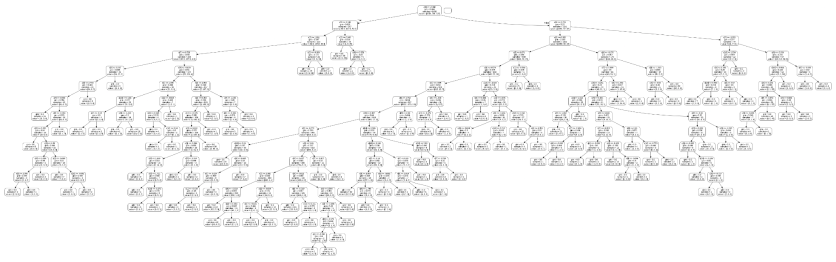


Model Performances

Random Forest

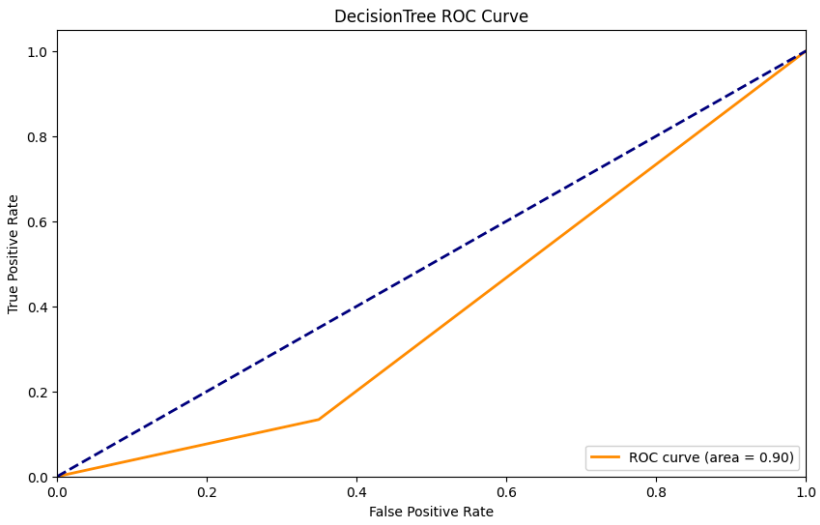
Different estimators	
Accuracy	
Precision and recall	
ROC AUC score and curve	

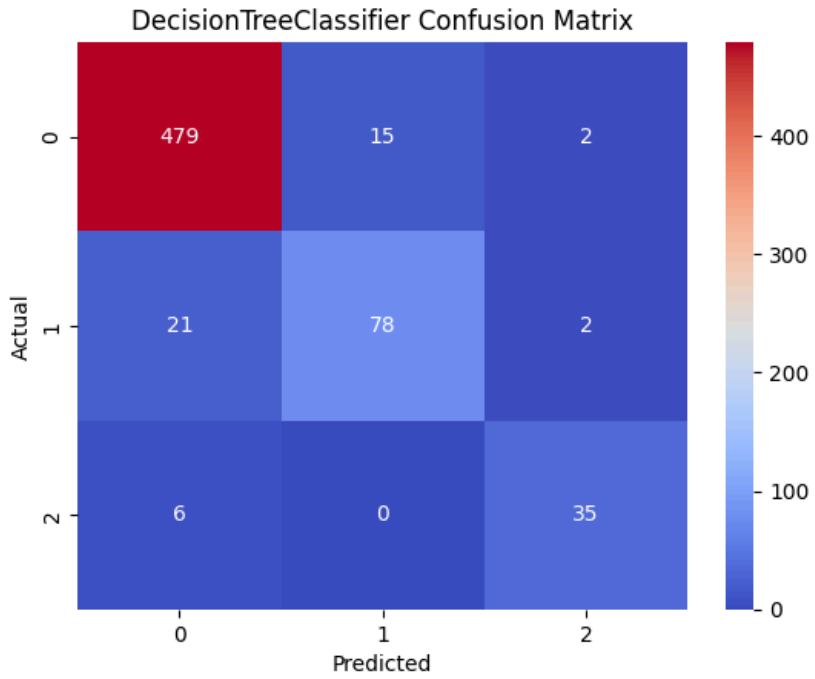
	
Confusion matrix	

Decision Tree	
Decision Tree Image	

Accuracy	`dtree_pred = dtree.predict(X_test)`
	
	`# Model accuracy`
	`dtree_acc = accuracy_score(y_test, dtree_pred)`
	
	`print("Accuracy Score: ", dtree_acc)`
	
	`Accuracy Score:  0.9294670846394985`

Precision and recall	`# Precision and Recall`
	`dtree_precision = precision_score(y_test, dtree_pred, average='weighted')`
	`dtree_recall = recall_score(y_test, dtree_pred, average='weighted')`
	
	`print("Precision Score: ", dtree_precision)`
	`print("Recall Score: ", dtree_recall)`
	
	`Precision Score:  0.928586974969042`
	`Recall Score:  0.9294670846394985`

ROC AUC score and curve	`# ROC AUC`
	`dtree_pred_proba = dtree.predict_proba(X_test)`
	`dtree_roc_auc = roc_auc_score(y_test, dtree_pred_proba, average='weighted', multi_class='ovr')`
	`y_test_binarized = label_binarize(y_test, classes=encoder.classes_)`
	`dtree_fpr, dtree_tpr, _ = roc_curve(y_test_binarized.ravel(), dtree_pred_proba.ravel())`
	
	`print(f"ROC AUC Score: {dtree_roc_auc}")`
	
	`ROC AUC Score: 0.8963141961545339`
	
	 

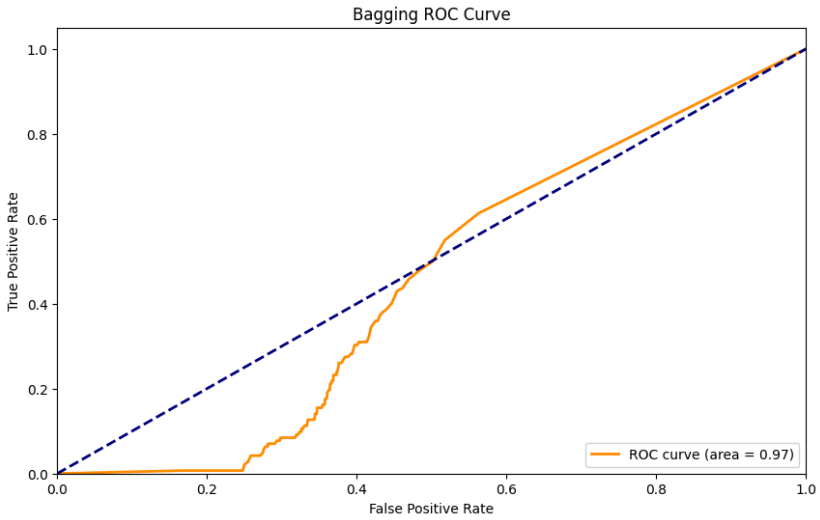
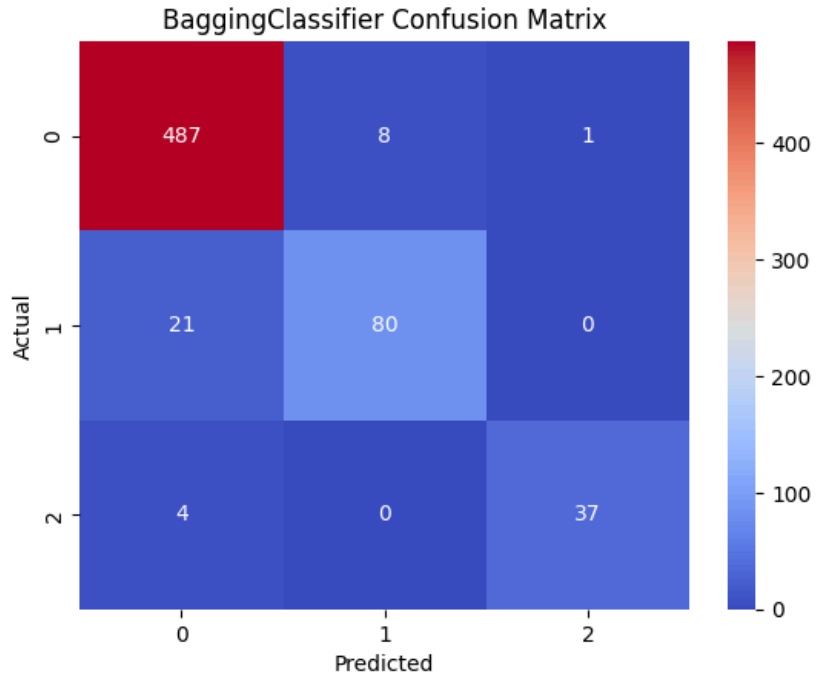
Confusion matrix	

Bagging	

Accuracy	```
	bag_pred = bag.predict(X_test)
	
	# Model Accuracy
	bag_acc = accuracy_score(y_test, bag_pred)
	
	print("Accuracy Score: ", bag_acc)
	``
	``
	Accuracy Score: 0.9467084639498433
	``

Precision and recall	```
	# Precision and Recall
	bag_precision = precision_score(y_test, bag_pred, average='weighted')
	bag_recall = recall_score(y_test, bag_pred, average='weighted')
	
	print("Precision Score: ", bag_precision)
	print("Recall Score: ", bag_recall)
	``
	``
	Precision Score: 0.9459568721783084
	Recall Score: 0.9467084639498433
	``

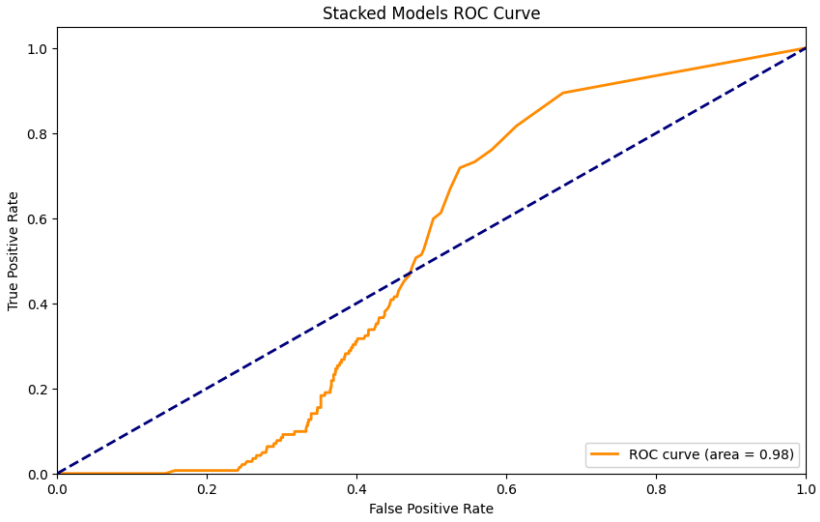
ROC AUC score and curve	```
	# ROC AUC
	bag_pred_proba = bag.predict_proba(X_test)
	bag_roc_auc = roc_auc_score(y_test, bag_pred_proba, average='weighted', multi_class='ovr')
	y_test_binarized = label_binarize(y_test, classes=encoder.classes_)
	bag_fpr, bag_tpr, _ = roc_curve(y_test_binarized.ravel(), bag_pred_proba.ravel())
	
	print(f"ROC AUC Score: {bag_roc_auc}")
	``
	``
	ROC AUC Score: 0.9723473876731273
	``

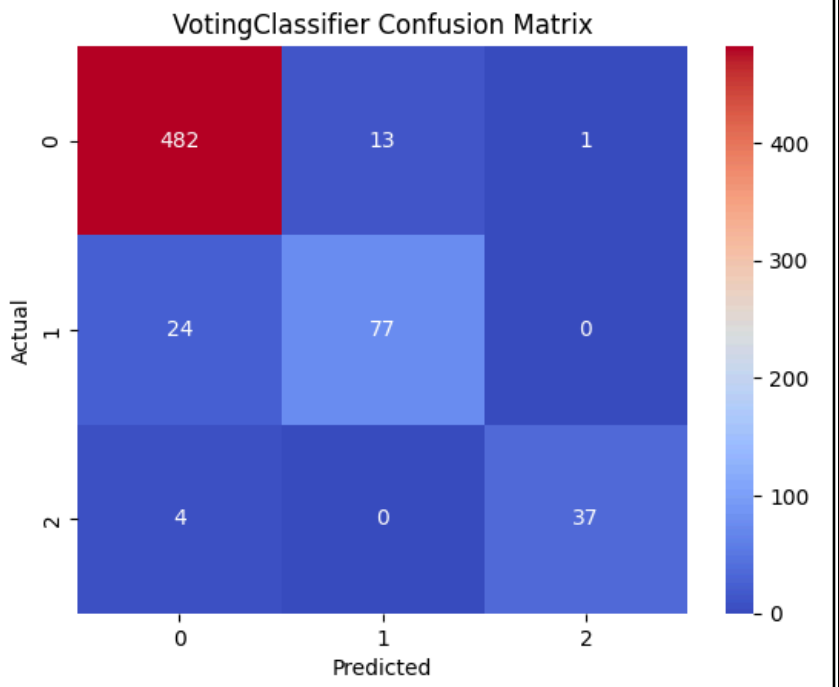
	
Confusion matrix	

Stacked Models	

Accuracy	```
	VC_pred = VC.predict(X_test)
	
	# Model Accuracy
	VC_acc = accuracy_score(y_test, VC_pred)
	
	print("Accuracy Score: ", VC_acc)
	``
	Accuracy Score: 0.9373040752351097

Precision and recall	```# Precision and Recall
	VC_precision = precision_score(y_test, VC_pred, average='weighted')
	VC_recall = recall_score(y_test, VC_pred, average='weighted')
	
	print("Precision Score: ", VC_precision)
	print("Recall Score: ", VC_recall)```
	``Precision Score: 0.9362294467771141
	Recall Score: 0.9373040752351097```

ROC AUC score and curve	```# ROC AUC
	VC_pred_proba = VC.predict_proba(X_test)
	VC_roc_auc = roc_auc_score(y_test, VC_pred_proba, average='weighted', multi_class='ovr')
	y_test_binarized = label_binarize(y_test, classes=encoder.classes_)
	VC_fpr, VC_tpr, _ = roc_curve(y_test_binarized.ravel(), VC_pred_proba.ravel())
	
	print(f"ROC AUC Score: {VC_roc_auc}")```
	```ROC AUC Score: 0.9753886599160393```
	 

Confusion matrix	

Google Colab link:
https://colab.research.google.com/drive/1-6t7Jc8lYeBYV8zztCmMl4u4UbgtGyR9?usp=sharing

5. Conclusion/Analysis/Learnings

Different machine learning models were used to predict fetal health, and they are evaluated based on their performances. These models, namely Random Forest, Decision Tree, Bagging, and Stacked Models, all returned positive model performances, ranging from 92 to 95 percent. The Bagging Classifier used 150 decision trees and returned 95% accuracy, precision, and recall scores and 97% ROC AUC score. This model performed the best among others, which can be considered as the most reliable and recommended model for predicting fecal health. The Random Forest classifier follows, which used 150 decision trees and showed 95% accuracy score, 94% precision score, 95% recall score, and 98% ROC AUC score The Stacked Models Classifier performed with 94% accuracy, precision, and recall scores, and 97% ROC AUC score. Lastly, the Decision Tree Classifier showed 92% accuracy, precision, and recall scores and 89% ROC AUC score, making it the least recommended in the following models.

The activity provided mastery of supervised machine learning models such as Decision trees, Random forest, Bagging method, and Stacking method as we are able to evaluate and compare their performances by calculating their accuracy, precision, recall, and ROC AUC score, as well as see their confusion matrices.

6. Assessment Rubric

Lab Activity Rubric

Criteria	Ratings						Pts
SO 7 PI 1 Student Outcome 7.1 Acquire and apply new knowledge from outside sources. threshold: 4.8 pts	6 pts Excellent | Educational interests and pursuits exist and flourish outside classroom requirements,knowledge and/or experiences are pursued independently and applies knowledge learned into practice	5 pts Good | Educational interests and pursuits exist and flourish outside classroom requirements,knowledge and/or experiences are pursued independently	4 pts Satisfactory | Look beyond classroom requirements, showing interest in pursuing knowledge independently	3 pts Unsatisfactory | Begins to look beyond classroom requirements, showing interest in pursuing knowledge independently	2 pts Poor | Relies on classroom instruction only	1 pts Very Poor | No initiative or interest in acquiring new knowledge	6 pts
SO 7 PI 2 Student Outcome 7.2 Learn independently threshold: 4.8 pts	6 pts Excellent | Completes an assigned task independently and practices continuous improvement	5 pts Good | Completes an assigned task without supervision or guidance	4 pts Satisfactory | Requires minimal guidance to complete an assigned task	3 pts Unsatisfactory | Requires detailed or step-by-step instructions to complete a task	2 pts Poor | Shows little interest to complete a task independently	1 pts Very Poor | No interest to complete a task independently	6 pts
SO 7 PI 3 Student Outcome 7.3 Critical thinking in the broadest context of technological change threshold: 4.8 pts	6 pts Excellent | Synthesizes and integrates information from a variety of sources; formulates a clear and precise perspective; draws appropriate conclusions	5 pts Good | Evaluate information from a variety of sources; formulates a clear and precise perspective.	4 pts Satisfactory | Analyze information from a variety of sources; formulates a clear and precise perspective.	3 pts Unsatisfactory | Apply the gathered information to formulate the problem	2 pts Poor | Gather and summarized the information from a variety of sources but failed to formulate the problem	1 pts Very Poor | Gather information from a variety of sources	6 pts
SO 7 PI 4 Student Outcome 7.4 Creativity and adaptability to new and emerging technologies threshold: 4.8 pts	6 pts Excellent | Ideas are combined in original and creative ways in line with the new and emerging technology trends to solve a problem or address an issue.	5 pts Good | Ideas are creative and adapt the new knowledge to solve a problem or address an issue	4 pts Satisfactory | Ideas are creative in solving a problem, or address an issue	3 pts Unsatisfactory | Shows some creative ways to solve the problem	2 pts Poor | Shows initiative and attempt to develop creative ideas to solve the problem	1 pts Very Poor | Ideas are copied or restated from the sources consulted	6 pts

Total Points: 24